

SWATHI K - Backend & Agentic AI Enthusiast

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Professional Summary

Backend-focused Software Engineer with hands-on experience building AI-powered applications, real-time systems, and scalable backend solutions using Java, Spring Boot, Python, FastAPI, React.js, PostgreSQL, Docker, and Redis. Experienced in developing AI agent workflows using OpenAI and LangGraph, REST APIs, authentication systems, and collaborative platforms. Strong foundation in DSA, OOP, DBMS, CN and backend architecture, with the ability to quickly learn and build reliable product-driven solutions..

Technical Projects

Multi-Agent Hackathon Assistant | React, Spring boot, postgresSQL, LangGraph, OpenAI API, Fast API [GitHub](#) | 06/2026

- Developed and orchestrated multiple AI agents using LangGraph for problem understanding, architecture generation, sprint planning, database design, and hackathon project guidance.
- Built agent communication workflows and shared state management to enable collaboration between specialized agents for project planning and execution.
- Integrated AI agent outputs with backend APIs to support intelligent features such as GitHub repository progress analysis, dynamic task tracking, risk detection, and hackathon judging simulation.
- Contributed to intelligent automation features including smart role assignment, project feasibility analysis, sprint optimization, and AI-driven PPT/demo preparation.

Real-Time Collaborative Code Editor | React.js, Spring Boot, WebSocket, PostgreSQL, Docker, OpenAI AP [GitHub](#) | 03/2026

- Built a real-time collaborative code editor with live cursor synchronization using WebSockets and STOMP.
- Implemented room-based collaboration with Admin, Editor, and Viewer role management.
- Designed isolated Docker-based code execution supporting Java, Python, Node.js, and C++ runtimes.
- Integrated OpenAI-powered code assistance, live chat, notifications, and JWT-based authentication.

Real-Time Sneaker Marketplace | React.js, Spring Boot, PostgreSQL, WebSocket [GitHub](#) | 01/2026

- Engineered a microservices-based sneaker marketplace with separate auth, product, order, and bidding services.
- Developed RESTful APIs to improve modularity, scalability, and independent service deployment.
- Deployed microservices on Kubernetes with auto-scaling and rolling updates for high availability.
- Implemented WebSocket-based live bidding, scaling to support over 200 concurrent participants with sub-second latency.

Internships Experience

- Web Development Intern, Learnloficify - Coimbatore** May – June 2025
 - Leveraged HTML, CSS, and JavaScript to create a responsive UI that reduced page load time by 20%.
 - Managed source code with Git and GitHub, enforcing branch protection and peer-review workflows.

Education Qualification

KIT - Kalaigarkarunanidhi Institute of Technology, Coimbatore 2023-2027
BTech - Computer Science and Business Systems GPA: 7.86

KIT - Kalaigarkarunanidhi Institute of Technology, Coimbatore 2022-2023
BTech - Computer Science and Business Systems PERCENTAGE: 67% - 403

Core Computer Science: Object Oriented Programming, Data BaseManagement Systems, Operating Systems, Computer Networks

Skills

Programming Languages: Java, Python, JavaScript, SQL

Frontend Technologies: HTML, CSS, React.js

Backend Technologies: Spring Boot, FastAPI, WebSocket

Databases: PostgreSQL

Agentic AI Technologies: LangGraph, OpenAI, ChromaDB

Tools & Platforms: Git, GitHub, Maven, Docker, Postman

Certifications

[NPTEL](#) (Java, C)

[Infosys Springboard](#) (HTML & CSS, Java)

[Coursera](#) (DSA)

[AWS](#) (Cloud Foundation)

Competitive Programming & Achievements

- Solved **900+ DSA problems** across LeetCode, CodeChef, Geekforgeeks, and Codeforces
- Leetcode** : Max rating - **1561** | Top **31.92%** globally | Badges : **7** | Highest Rank : **5035**
- Codechef** : Max rating - **1512** | Position : **2** Star | Highest rank **126** in Starters 221 (Rated) | Global Rank : **1129**
- participated multiple hackathons (3hrs Agentic AI, 12-hours and 24-hours DSA hackathon)